

RATIONAL EXPRESSIONS AND EQUATIONS (Chapter 4)

LESSON 1 - Factoring Review

VOCABULARY

Factoring - Factoring means to write a polynomial as the product of simpler polynomials.

- Factoring is a key skill for rational functions and equations.

Greatest Common Factor

- Always look for this first!
- Find the GCF of both terms and place it outside the brackets.
- Divide each term inside the brackets by the GCF and write the result inside the brackets.
- Pay attention to the signs!



The graphing calculator can help you find the **GCF** of two numbers.

- Math → Num → GCD (9)

Enter the two **positive** numbers separated by a comma (above the number 7).

Example:



- The GCF of two variables is the common variable with the smallest exponent.

Example: $\therefore gcf(x^3, x^2) = x^2$

EXAMPLE 1

Factor the following completely.

a) $10x^2 - 15x$

b) $3x^4 + 12x$

c) $-16x^3 - 24x^2$

Difference of Squares $a^2 - b^2$

- Difference of squares are two terms subtracted; no middle term.
- Place the square root of the first number as the first term in each bracket, the square root of the second number as the second term in each bracket and separate them by + and – in any order.

$$\begin{array}{ccc} & a^2 - b^2 & \\ \swarrow & & \searrow \\ \sqrt{a^2} = a & & \sqrt{b^2} = b \\ \text{first term} & & \text{second term} \\ & (a + b)(a - b) & \end{array}$$

EXAMPLE 2

Factor the following completely. Always look for GCF first!

a) $x^2 - 49$

b) $4x^2 - 9$

c) $20x^2 - 5$

Leading Coefficient

- The number in front of the variable term with the highest exponent is the leading coefficient.
- When factoring you want to leave the resulting polynomial with **positive** leading coefficient.

EXAMPLE 3

Factor so that the leading coefficient is positive.

a) $3 - x$

b) $-20x^3 - 45x$

c) $18x - 27x^3$

Practice

1. Factor the following completely.

a) $2x^4 + 8x$	b) $2x^2 - 12x$
c) $3x^2 - 9x$	d) $-32x^2 - 24x$
e) $10x^6 + 2x^2$	f) $7x^3 + 35x^5$

2. Factor the following completely. Always look for GCF first!

a) $x^2 - 16$	b) $x^2 - 1$
c) $4x^2 - 9$	d) $4x^3 - x$

3. Factor so that the leading coefficient is positive.

a) $4 - x$	b) $3x - 9x^2$
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c) $-15x^3 - 5x$	d) $9 - x^2$
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4. Factor the following completely. Always look for GCF first!

a) $9x^2 - 1$	b) $4x - 36$
c) $x^2 - x$	d) $64x^2 - x$
e) $144x - 4x^3$	f) $4x^2 + 36$
g) $-64x + 4x^2$	h) $108 - 3x^2$
i) $36x^2 + 16x^3$	j) $x^2 - 64$

LESSON 2 - Equivalent Expressions & Non-permissible values (NPV's)

VOCABULARY

Rational Expression - A fraction that has a polynomial for both the numerator and the denominator.

$$\text{Ex. } \frac{2x}{3x+4}$$

Non-permissible value - The value of a variable that makes the denominator of a rational expression equal to zero.

For example, $\frac{10}{15+x}$ is undefined when $x = -15$, because this creates a zero denominator.

Therefore, $x = -15$ is a non-permissible value for $\frac{10}{15+x}$.

In this unit, we will learn how to work with rational expressions, a fancy phrase for 'fractions with variables'. We will revisit what we learned way back in junior high: how to add, subtract, multiply and divide fractions, but our fractions will have polynomials in the numerator and denominator. Let's remember some basic fraction rules first:

Consider the fractions $\frac{4}{8}$ and $\frac{12}{24}$.

- What can you say about this pair of fractions?
- What is another equivalent fraction to the pair above?

Let's look at a more complicated rational expression: $\frac{6x}{9y}$.

- Find three equivalent rational expressions.

So, to create any equivalent rational expression, we simply _____ or _____ both the _____ and the _____ by the same amount. This amount could be a number or a _____.

EXAMPLE 1

Determine an equivalent rational expression for $\frac{4x^2+8x}{4x}$ by:

a) multiplying

b) factoring then dividing

EXAMPLE 2

Determine if the following rational expressions are equivalent:

a) $\frac{9}{3x-1}$ and $\frac{-18x}{2-6x}$

b) $\frac{2-2x}{4x}$ and $\frac{x-1}{2x}$

Non-Permissible Values

- It is “OK” for the numerator to be zero.
- A zero in the denominator is a big “NO”!
- NPVs are the values of the variable that make the denominator equal to zero.
- To algebraically find the NPVs, factor the denominator, set each factor to zero and solve for the variable.
- The NPVs of a rational function are the x -intercepts of the denominator.
- When dealing with rational expressions and equations, state the NPV as restrictions.



The highest exponent of the variable tells you the maximum number of NPVs.

EXAMPLE 3

State the non-permissible values and restrictions of the following.

a) $\frac{4x+3}{x}$

b) $\frac{5x}{4xy}$

c) $\frac{x}{x(2x-3)}$

d) $\frac{x+3}{5x^2-x}$



Monomial denominators always have restrictions variable $\neq 0$.

EXAMPLE 4

a) Write a rational expression that has a non-permissible value of $x = 0$.

b) Write a rational expression that has non-permissible values of $x = -4$ and $x = \frac{2}{3}$.

Summary

- Rational expressions are nothing more than fractions, but with _____ in the numerator and denominator.
- We can find equivalent fractions by _____ or _____ the numerator and denominator by the same amount.
- When polynomials are involved we may have to simplify by _____ and then dividing common factors, to determine if expressions are equivalent.
- Since division by zero is not defined, values of a variable that make the denominator equal to zero are not permitted; that is, variables have restrictions. We must always find these, even if it is not explicitly stated in the question!

Practice

1. Identify the rational expressions that are equivalent to $\frac{48x^2}{16x-4}$.

A. $\frac{12x^2}{4x-1}$

B. $\frac{24x^2}{8x-4}$

C. $\frac{3x^2}{x-4}$

D. $\frac{36x^2}{48x-12}$

2. State the non-permissible values for each expression.

a) $\frac{16x^3}{10x}$	b) $\frac{4x^2}{x^2+x}$	c) $\frac{2x+6}{4}$
d) $\frac{x-2}{4x^2}$	e) $\frac{x}{x-1}$	f) $\frac{2-7x}{-5-x}$
g) $\frac{5x}{(x-8)(x+2)}$	h) $\frac{-1}{15x^2+10x}$	h) $\frac{2x-2}{4x^2-4}$
i) $\frac{3x^2}{x^2+9}$	j) $\frac{x-2}{2x^2-8}$	k) $\frac{1-x}{2x(x-1)}$

3. State the non-permissible values of the variable in each rational expression below, and write two equivalent expressions.

Rational expression	Non-permissible value(s)	Equivalent expression #1	Equivalent expression #2
a) $\frac{-x}{3x^2-12}$			
b) $\frac{-15-6y}{3y^3}$			

4. Write a rational expression that has non-permissible values of $x = 0$ and $x = 3$.

5. Write a rational expression that has non-permissible values of $x = -5$ and $x = \frac{1}{2}$.

LESSON 3 - Simplifying Rational Expressions

STEPS:

- **Factor** numerator and denominator completely (positive leading coefficient).
- State **non-permissible values** as restrictions BEFORE simplifying.
- **Simplify** by removing common factors (not common terms).
Divide out the common factors in the numerator and denominator.



- Same terms, same sign simplifies to **+1**

$$\frac{a+b}{b+a} = 1$$

$$\text{ie. } \frac{x+1}{1+x} = 1$$

- Same terms, opposite sign simplifies to **-1**

$$\frac{a-b}{b-a} = -1$$

$$\text{ie. } \frac{3-x}{x-3} = -1$$

EXAMPLE 1

Simplify the following rational expressions and state all non-permissible values as restrictions:

a) $\frac{14x^5}{4x^2}$	b) $\frac{-24x^2}{18x^3}$
c) $\frac{15x^3-5x}{15x^3}$	d) $\frac{3x-9x^2}{6x^3}$
e) $\frac{6x^2-8x}{3x^3-4x^2}$	f) $\frac{2x-1}{4x^2-1}$
g) $\frac{1-x}{x^2-1}$	h) $\frac{2c-8}{16-4c}$

Practice

1. Simplify the following rational expressions and state the non-permissible values as restrictions.

a) $\frac{6x^2+24x}{6x}$	b) $\frac{14a^2}{21a-28a^2}$
c) $\frac{3x^2+15x}{6x^2-3x}$	d) $\frac{9m^3+15m^4}{6m^2}$
e) $\frac{x^2-100}{(x+10)^2}$	f) $\frac{12a^2-8a^3}{6x^2-3x}$
g) $\frac{1-x}{x-1}$	h) $\frac{y+9}{y^2-81}$

2. Given that $\frac{ax-b}{cx^2-6}$ is equivalent to $\frac{2x-4}{6x^2-12}$, the values of a , b , and c , are, respectively, _____, _____, _____.

Use the following information to answer the next question.

Sidney tried to simplify an expression, as shown below.

$$\begin{aligned}\frac{8-2x^2}{2x^2+4x} &= \frac{2(4-x^2)}{2x(x+2)} \\ &= \frac{2(2-x)(2+x)}{2x(x+2)} \\ &= \frac{2-x}{2x}, x \neq 0\end{aligned}$$

3. Identify Sidney's error and correct the solution. **Explain** why Sidney made this error and correct her error.

Use the following information to answer the next question.

A farmer is planning to put a fence around a rectangular field. The area of the field, in square meters, can be represented by the expression $18a^2 - 2$ and the width of the field is $(3a + 1)$, as shown below.

$$A = 18a^2 - 2$$

$$(3a + 1)$$

4. a) Write and simplify an expression for the length of the field.

- b) If the farmer requires 88 m of fencing, **determine** the value of a and the length and the width of the field.

LESSON 4 - Multiplying and Dividing Rational Expressions

Multiplying/Dividing Rational Expressions

Let's see if we remember how to multiply and divide fractions, without our calculator!

Multiply and divide the following fractions:

$$\frac{3}{2} \cdot \frac{2}{9} =$$

$$\frac{1}{3} \div \frac{7}{6} =$$

Now, let's try it with rational expressions.

EXAMPLE 1

Multiply and divide the following, state all restrictions.

a) $\frac{5x^2}{2} \cdot \frac{3}{7x}$	b) $\frac{10}{3x} \cdot \frac{6x}{5x^3}$
c) $\frac{2x^2-12x}{15x} \cdot \frac{5x}{x-6}$	d) $\frac{x-5}{3x^2-9x} \div \frac{5}{6x-18}$
e) $\frac{x-1}{x+4} \div \frac{3x-3}{x^2-16}$	f) $\frac{9x^2+36x}{x^2-16} \cdot \frac{6x+4}{6x-30} \div \frac{6x^2+4x}{3x-12}$

EXAMPLE 2

The base of a triangle is given by $\frac{4x-16}{x}$ and the height of the triangle is $\frac{x}{x^2-16}$. The simplified form of the area of the triangle can be written in the form of $\frac{A}{Bx+C}$. **Determine** the values of A , B , and C .

Summary

Key Idea

- The strategies you use to multiply and divide rational numbers can be used to multiply and divide rational expressions. That is, if A , B , C , and D are polynomials, then

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}, B, D \neq 0$$

$$\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C} \text{ or } \frac{AD}{BC}, B, D, \text{ and } C \neq 0$$

- If we have a binomial, try _____ first, then try simplifying before multiplying/dividing.
- Identify the _____ of the variables.
- When multiplying fractions, multiply the numerator with _____ and denominator with _____.
- When dividing fractions, take the _____ of the second fraction and then _____.
- When dividing, we must look for restrictions in _____ places, because we are going to take the reciprocal of the second fraction.

- Simplify using _____ factors.

Practice

1. State all non-permissible values for the following rational expressions.

a) $\frac{4x}{(x-3)} \cdot \frac{2x(x+2)}{(3x-1)}$

b) $\frac{8x^3}{(x+4)} \div \frac{x}{(x-1)}$

c) $\frac{4x}{7} \cdot \frac{2x(2x+5)}{12}$

d) $\frac{3}{x-1} \div \frac{x+3}{x}$

Use the following information to answer the next question.

Aaron is working on simplifying the expression $\frac{2x^2-18}{12-6x} \div \frac{2x+10}{x^2-4}$.

Six statements are made regarding the non-permissible values of this expression.

Statement 1 $x = 3$

Statement 2 $x = -3$

Statement 3 $x = 2$

Statement 4 $x = -2$

Statement 5 $x = 5$

Statement 6 $x = -5$

2. The three statements above that represent the non-permissible values of the expression are statements _____, _____, and _____.

3. Simplify the following expressions and state the non-permissible values as restrictions.

a) $\frac{4x^2}{27} \cdot \frac{9}{2x^3}$

b) $\left(\frac{5x^3}{12x}\right)\left(\frac{5}{15x^3}\right)$

c) $\frac{-6x^2}{18} \div \frac{3x^3}{14}$

c) $\frac{2m^2n^3}{6p} \cdot \frac{9m^3p^5}{2n^4}$

4. The number of distinct non-permissible values in the expression $\frac{(x-1)(x+7)}{(x+7)(x-9)} \div \frac{x}{(x+8)(x-9)}$ is

A. 5

B. 4

C. 3

D. 2

5. Simplify the following expressions and state the non-permissible values as restrictions.

a) $\frac{x^2+3x}{25} \cdot \frac{15x^3}{9x+3x^2}$

b) $\frac{x^4}{5x^2-2x^3} \cdot \frac{2x^3-5x^2}{x^3}$

c) $\frac{x+2x^2}{1-x} \cdot \frac{x^2-x}{2x^4+x^3}$

c) $\frac{x^2-9}{3x+9} \cdot \frac{15x^3}{12x^2-36x}$

6. Simplify the following expressions and state the non-permissible values as restrictions.

a) $\frac{6-x}{x-1} \div \frac{4x+4}{2x-2x^2}$

b) $\frac{4x^2+6x}{10x^3} \div \frac{2x+3}{15}$

c) $\frac{25x^3}{10x^3-20x^2} \div \frac{-x}{4x-8}$

c) $\frac{x^4+6x^3}{18x^2+6x} \div \frac{-4x^2-24x}{10x^3+3x^4}$

7. Simplify the following expressions and state the non-permissible values as restrictions.

a) $\left(\frac{49m^4n^6}{18p^3m^2}\right)\left(\frac{4mp}{14n^2}\right)$

b) $\left(\frac{12x^4y^6}{18yz^2}\right) \div \left(\frac{x^4z^6}{12y^2}\right)$

8. When the function $\frac{6x-18}{18(x^2-9)}$ is simplified to the form of $\frac{A}{Bx^C+D}$, the values of A, B, C, D are

_____ , _____ , _____ , _____
 $A \quad B \quad C \quad D$

9. The non-permissible values of x in the expression $\frac{x-2}{x+2} \cdot \frac{x^2-9}{x+3}$ are

- A. $-3, -2, 2, 3$
- B. $-2, 2, 3$
- C. $-3, -2, 3$
- D. $-3, -2$

LESSON 5 - Adding and Subtracting Rational Expressions

Adding/Subtracting Rational Expressions

Let's see if we remember how to add and subtract fractions, without our calculator!

Simplify the following expressions:

$$\frac{2}{3} + \frac{5}{4} =$$

$$\frac{4}{7} - \frac{3}{8} =$$

So, adding/subtracting fractions is slightly more difficult than multiplying/dividing, because we must find a common _____. Now, if we can determine the lowest common denominator that is even better!



Incredibly important! Only **factors** of the numerator and the denominator that are the same may be divided!

You May Divide	You May NOT Divide
$\frac{6}{2} = \frac{2(3)}{2}$	$\frac{2-1}{4}$
$\frac{x(x+1)}{(x+1)}$	$\frac{(x-1)+4}{4}$
$\frac{(x^2-9)(x+3)}{(x^2-9)}$	$\frac{(x-6)-(x+2)}{(x+2)}$
$\frac{(\square)(\square)}{(\square)}$	$\frac{(\square) + (\triangle)}{(\square)}$

Now, let's try it with rational expressions.

EXAMPLE 1

Add/Subtract the following, state all restrictions.

a) $\frac{3}{8x^2} + \frac{1}{4x}$	b) $\frac{3x}{2x+1} - \frac{4}{x-3}$
c) $\frac{2}{x-2} - \frac{3}{x}$	d) $\frac{x}{x-2} - \frac{x-5}{x^2-4}$
e) $\frac{32}{x^2-16} + \frac{4}{x-4}$	f) $\frac{x^2+9}{9-x^2} - \frac{x}{3-x}$

Summary

- _____ completely.
- State all _____.
- Simplify if possible.
- To add/subtract fractions, we must find a
_____.
- Multiply the numerator and denominator of each term by the missing factors from the
_____.

- When subtracting, be extra careful with the

Practice

1. State the lowest common denominator for each pair of rational expressions.

a) $\frac{2x}{5x^2}$ and $\frac{3y}{10x}$ b) $\frac{-1}{6x^3}$ and $\frac{4y}{9x}$ c) $\frac{2x}{x(x+1)}$ and $\frac{4}{(x-1)}$ d) $\frac{13}{(2x-1)(x+1)}$ and $\frac{-1}{(2x-1)(x-1)}$

2. State the non-permissible values for each pair of rational expressions in 1.

a) b) c) d)

3. Simplify the following expressions and state all restrictions.

a) $\frac{12x^2}{5x} - \frac{6x^2}{5x}$	b) $\frac{5x-1}{x-1} - \frac{3x+1}{x-1}$
c) $\frac{1}{x^2+2x} - \frac{5x+3}{x^2+2x}$	d) $\frac{-6x+5}{4x^2} + \frac{3}{6x}$
e) $\frac{3x-7}{6x} - \frac{1}{12x^2}$	f) $\frac{5x+12}{x(x+6)} - \frac{3}{x}$

g) $\frac{-2}{2x-3} + \frac{1}{4(x+1)}$	h) $\frac{4(x-6)}{x^2-36} - \frac{18+3x}{6(x-6)}$
i) $\frac{6}{x^2-1} + \frac{1}{(x+1)}$	j) $\frac{3x}{2x+1} - \frac{3}{4x^2-1}$

4. The expression $\frac{2}{x} + \frac{x+4}{x^2-3x}$, where $x \neq 0, 3$, is equivalent to

A. $\frac{x+6}{x^2-3x}$

B. $\frac{2x-2}{x^2-3x}$

C. $\frac{3x+1}{x^2-3x}$

D. $\frac{3x-2}{x^2-3x}$

Use the following to answer the next question.

The simplified sum of $\frac{x+4}{3x} + \frac{1}{2}$, $x \neq 0$ can be written in the form

$$\frac{Ax+B}{Cx}$$

where A, B, and C represent single-digit numbers.

5. In the simplified sum $\frac{Ax+B}{Cx}$, the value of

A is _____.

B is _____.

C is _____.

LESSON 6 - Solving Rational Equations

VOCABULARY

rational equation - An equation that involves one or more rational expressions Ex. $\frac{5}{x} = \frac{4}{x+2}$

Now, we will work with equations. The difference between **an equation** and **an expression** is simply an _____.

Solving rational equations:

- Factor completely (if possible) and state NPV's
- Determine the common denominator, and
- Multiply all terms in the equation by the common denominator.
- The equation will result in a linear or quadratic equation that needs to be solved *If the equation is not linear or quadratic look for a GCF that can be factored.*
 - o **LINEAR**: Variable on one side of the equation and number on the other.
 - o **QUADRATIC**: Use inverse operations to get the entire equation on one side and zero on the other $ax^2 + bx + c = 0$. Use the quadratic formula or factoring to solve for x.
 - $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ (on formula sheet)

REVIEW QUESTIONS (Solving linear and quadratic equations)

Solve the following equations.

a) $9(x + 2) = 4(3x)$	b) $(x + 7) + 3(2x) = -2(x + 7)$
c) $2x + (x - 3) = x(x - 1)$	d) $(x - 3)(x + 1) = 3x(x + 5)$

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ALWAYS check your answer for any equation you solve! Either substitute the value into the original equation to see if LS = RS or use your calculator to do the check.

Extraneous Roots

- Solutions to the linear or quadratic equation that are **NOT solutions** to the original equation.
- Always check solutions against the NPV and in the original equation.
- Pay attention to extraneous roots when solving word problems.
 - Time, speed, distances, etc.... can't be negative.
 - Number of objects and people can't be decimals.

1. Fraction: Cross products

- Multiply the numerator of Fraction 1 by the denominator of Fraction 2.
- Multiply the denominator of Fraction 1 by the numerator of Fraction 2.
- Set the products equal and solve.

$$\frac{a}{b} = \frac{c}{d}$$

$$\frac{a}{b} \swarrow \searrow \frac{c}{d}$$

$$a \times d = b \times c$$

EXAMPLE 1

Solve the following equations. State any restrictions.

a) $\frac{5x-4}{3x-1} = \frac{3}{2}$	b) $\frac{x+4}{6x} = \frac{4}{3x}$
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c) $\frac{8}{3x-2} = \frac{2}{x-1}$	d) $\frac{3}{x-5} = \frac{10}{5x-25}$

2. Fraction $\neq$ Fraction: Eliminate the denominators

- Factor the denominators, find the LCM and state NPV as restrictions.
- Multiply each term of the equation by the LCM to eliminate all the fractions.
- Solve the resulting linear or quadratic equation and check the solution(s).

EXAMPLE 2

Solve the following equations. State any restrictions.

a) $\frac{3x}{4} - \frac{5}{3} = \frac{7}{2}$	b) $\frac{10}{x} + 3 = \frac{x+9}{x-4}$
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c) $\frac{18}{x^2-3x} = \frac{6}{x-3} - \frac{5}{x}$	d) $\frac{2x}{x+3} + \frac{x}{x-3} = \frac{18}{x^2-9}$

Summary

- To solve a rational equation, first _____ everything.
- Find a _____ and multiply all terms by it. You should not have fractions after this.
- The solve the resulting _____ or _____ equation.
- Remember, we must always check the solution with the _____.
If our solution is one of our non-permissible values, then we have an _____.
- We can always check our work by _____.
- If we end up with any term bigger than x^2 such as x^3 or x^4 , we did **not** make the simplest common

denominator. Go back and check.

- If you really want to “cross-multiply”, this can **ONLY** be done when there is only _____ term on both sides of the equal sign.

Practice

1. Solve each rational equation below. **Hint:** Always state your NPVs even if not asked and verify your solution.

a) $\frac{1}{x} + \frac{1}{3x} = \frac{1}{6}$

b) $\frac{1}{x} + \frac{1}{x+1} = \frac{1}{6}$

c) $\frac{2x-6}{4x-8} = 1$

d) $\frac{3}{x-5} = \frac{10}{5x-25}$

e) $\frac{3}{2x} - \frac{x-1}{4x} = 1$

f) $\frac{4x-1}{2x} + \frac{2x+3}{3x^2} = 2$

g) $\frac{x+3}{x-1} = 2x + 1$	h) $\frac{3}{3x+2} = \frac{6}{5x}$
i) $\frac{1}{x-3} - \frac{6}{x^2-9} = \frac{1}{5}$	f) $\frac{x}{x-1} - \frac{1}{x-1} = -4$

Use the following information to answer the next question.

The rational equation $\frac{2+x}{x+2} = \frac{3}{x}$ is solved for x using the work shown below.

$$x(2 + x) = 3(x + 2)$$

$$x^2 + 2x = 3x + 6$$

$$\begin{aligned}
 x^2 - x - 6 &= 0 \\
 (x - 3)(x + 2) &= 0 \\
 x = 3 \text{ or } x &= -2
 \end{aligned}$$

2. Which statement correctly completes the solution?

- A. Both $x = 3$ and $x = -2$ are valid solutions.
- B. Both $x = 3$ and $x = -2$ are extraneous solutions.
- C. The solution $x = 3$ is a valid solution, but $x = -2$ is an extraneous solution.
- D. The solution $x = -2$ is a valid solution, but $x = 3$ is an extraneous solution.

Use the following information to answer the next question.

When solving the equation $\frac{3}{x-2} + \frac{5x}{10-5x} = 2$, $x \neq 2$, Darren made an error. His work is shown below.

Step I $\frac{3}{x-2} + \frac{5x}{5(2-x)} = 2$

Step II $\frac{3}{x-2} + \frac{x}{(x-2)} = 2$

Step III $\frac{3(x-2)}{x-2} + \frac{x(x-2)}{(x-2)} = 2(x-2)$

Step IV $3 + x = 2x - 2$

Step V $x = 5$

3. The first recorded error in Darren's solution is in Step _____.

4. **Algebraically determine** the correct solution to Darren's equation: $\frac{3}{x-2} + \frac{5x}{10-5x} = 2$, $x \neq 2$.

LESSON 7 - Applications of Rational Equations

Word Problems

- Identify and define the unknown.
- Use a table (Distance Speed Time or Job Share questions) to organize the information given in the question.
- State the equation and solve it algebraically (written response) or using your graphing calculator (multiple choice or numeric response).
- Rule out inadmissible solutions:
 - o Can't have negative values for time, distance, etc.
 - o Can't have decimal values for the number of items.
 - o Can't have solutions that match the NPV.

EXAMPLE 1

Marc drove 404 km from Edmonton to Banff in the same length of time Amber took to drive 364 km from Edmonton to Jasper. Marc drove 10 km/h faster than Amber. **Determine** Marc's speed, in kilometers per hour.

	Distance (km)	Speed (km/h)	Time (h)
Marc			
Amber			

EXAMPLE 2

When they work together, Stuart and Lucy can deliver flyers to all the homes in their neighbourhood in 42 mins. When Lucy works alone, she can deliver the flyers in 13 minutes less than Stuart can when he works alone. When Stuart works alone, **determine** how long he takes to deliver the flyers.

	Time (min)	Fraction of time to complete in 1 min
Stuart		
Lucy		
Together		

EXAMPLE 3

Steve bought a case of vintage hockey cards for \$450. He kept two for himself and sold the rest for \$560, making a profit of \$10 for each hockey card, represented by the equation below:

$$\frac{560}{v-2} - \frac{450}{v} = 10, \quad v > 2, \quad v \in W$$

where v represents the number of vintage hockeycards in the case. **Determine** the number of vintage hockey cards in the case.

Practice

Use the following information to answer the next question.

Myha and Liam traveled 264 km by car from Calgary to Edmonton to watch a football game. On the way to the game, Myha drove. On the return trip, Liam drove 10 km/h faster. By doing so, he saved 0.3 hours of traveling time. The partially completed table below shows this information.

	Distance (km)	Speed (km/h)	Time (h)
Myha	264		
Liam	264	x	

1. The equation that can be used to solve for Liam's speed on the return trip is

A. $\frac{264}{x} = 0.3 + \frac{264}{x-10}$

B. $\frac{264}{x-10} = 0.3 + \frac{264}{x}$

C. $\frac{264}{x+10} = 0.3 - \frac{264}{x}$

D. $\frac{264}{x} = 0.3 - \frac{264}{x+10}$

2. Ted is a long-distance driver. It took him 30 min longer to drive 275 km on the Trans-Canada Highway west of Swift Current, Saskatchewan, than it took him to drive 300 km east of Swift Current. He averaged 10 km/h less while travelling west of Swift Current due to more severe snow conditions. **Determine** Ted's average speed for each part of the trip.

	Distance (km)	Speed (km/h)	Time (h)
East			
West			

2. When they work alone, it takes Jill 3h to paint a room and Jack 2h to paint a room that is the same size. **Determine** how long it would take them if they worked together.

	Time (h)	Fraction of time to complete in 1 hour
Jill		
Jack		
Together		

Use the following information to answer the next question.

The average speed of an airplane is eight times as fast as the average speed of a train. To travel 1200 km, the train requires 14 more hours than the airplane does. This can modeled by the equation

$$\frac{1200}{x} - \frac{1200}{8x} = 14$$

where x is the average speed of the train in km/h.

3. **Algebraically determine** the average speed of the airplane and the train.

Use the following information to answer the next question.

Suppose you are running water into a tub. The tub can be filled in 2 minutes if only the cold water is used. It fills in 3 minutes if only the hot water is used. This situation can be modeled by the equation,

$$\frac{t}{2} + \frac{t}{3} = 1$$

where t represents time in minutes.

4. **Determine** how long will it take to fill the tub if both taps are on.

5. James solved the following rational equation.

$$\frac{3y}{y-1} + 6 = \frac{6y-9}{y-1}$$

He claims that the solution is $y = 1$. **Explain** why James's solution is incorrect. **Algebraically determine** the correct solution.

Use the following information to answer the next question.

When a scuba diver is below 33 ft, the time, t , in minutes, that she can remain underwater and then safely surface without a decompression stop can be modeled by the function

$$t = \frac{1700}{d-33}$$

where d represents the maximum depth of the dive in feet.

6. During a 50-minute dive, the maximum depth, to the nearest foot, to which the diver can dive and then safely surface without a decompression stop is _____.